

INDIAN KNOWLEDGE SYSTEM PROJECT

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Ayurvedic Classification – Rule-Based Expert System

Title: Ayurvedic Classification as a Rule-Based Expert System

Problem Statement

Ayurveda is a traditional Indian knowledge system that classifies individuals and health-related conditions using concepts such as **Vata, Pitta, and Kapha** (the three doshas). In this project, the Ayurvedic classification concept is mapped to a **Rule-Based Expert System** in Computer Science.

The system accepts simple user characteristics such as **body tendency, appetite, activity level, sleep pattern, and response to weather**. It applies predefined **IF–THEN rules** and calculates which dosha has the highest score.

The objective is to demonstrate how traditional knowledge can be represented computationally using **facts, rules, inference, and classification**.

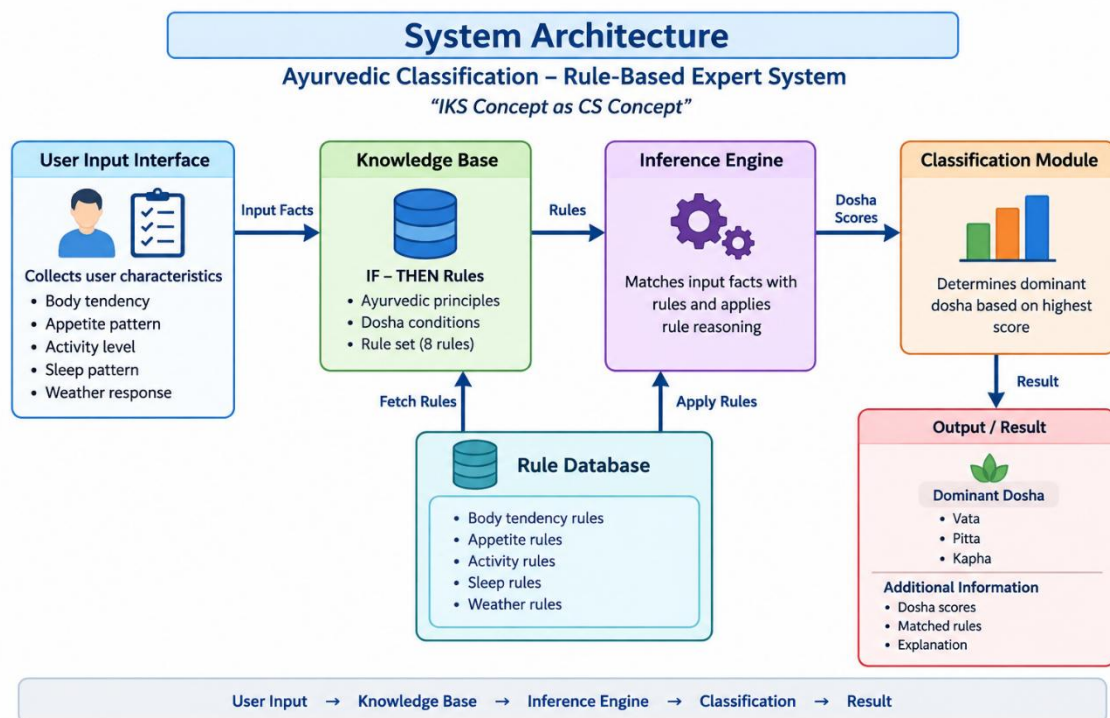
Conceptual Mapping Table

Ayurvedic Classification – Rule-Based Expert System

Sr. No.	Ayurvedic / IKS Concept	Computer Science Concept	System Implementation
1	Vata, Pitta, Kapha	Classes / Categories	Three classification categories
2	Dosha characteristics	Features / Attributes	Body, appetite, activity, sleep, weather
3	Ayurvedic principles	Knowledge Base	Stored as predefined rules
4	Traditional expert knowledge	Knowledge Representation	IF–THEN production rules
5	Ayurvedic classification rules	Production Rules	Example: IF appetite = strong THEN Pitta
6	Observation of characteristics	User Input / Facts	User selects characteristics
7	Applying Ayurvedic rules	Inference Engine	Python checks matching rules
8	Dosha assessment	Classification	Highest score determines result
9	Dominant Dosha	Predicted Class	Vata / Pitta / Kapha
10	Multiple observations	Feature Set	Collection of user inputs
11	Rule matching	Pattern Matching	Input values are compared with rules
12	Dosha score	Classification Score	Each matched rule adds one point

Sr. No.	Ayurvedic / IKS Concept	Computer Science Concept	System Implementation
13	Expert reasoning	Rule-Based Reasoning	IF–THEN logical inference
14	Traditional knowledge source	Database / Rule Repository	Rule dictionary in Python
15	Final Ayurvedic classification	System Output	Displays dosha and scores

Basic Architecture



Algorithm: Ayurvedic Rule-Based Classification

Objective

To classify a user's characteristics into **Vata, Pitta, or Kapha** using predefined **IF–THEN** rules.

Input

- Body tendency
- Appetite pattern
- Activity level
- Sleep pattern
- Weather response

Output

- Vata score
- Pitta score
- Kapha score
- Dominant dosha

Step-by-Step Algorithm

1. **Start**
2. Initialize three scores:
 - Vata = 0
 - Pitta = 0
 - Kapha = 0
3. Ask the user to enter their **body tendency**.
4. Apply the corresponding rule:
 - IF body = thin → increase Vata
 - IF body = medium → increase Pitta
 - IF body = solid → increase Kapha
5. Ask for the **appetite pattern**.
6. Apply the corresponding IF–THEN rule.
7. Ask for the **activity level** and apply its rule.
8. Ask for the **sleep pattern** and apply its rule.
9. Ask for the **weather response** and apply its rule.
10. Add the scores obtained from all matching rules.
11. Compare the three scores.
12. If Vata has the highest score, classify as **Vata**.
13. Else if Pitta has the highest score, classify as **Pitta**.
14. Else if Kapha has the highest score, classify as **Kapha**.
15. If two or more scores are equal and highest, classify as **Mixed/Equal**.
16. Display the scores, matched rules, and final classification.
17. **Stop**.

Pseudocode

```
START
Vata ← 0
Pitta ← 0
Kapha ← 0

INPUT body
IF body = "thin"
    Vata ← Vata + 1
ELSE IF body = "medium"
    Pitta ← Pitta + 1
ELSE IF body = "solid"
    Kapha ← Kapha + 1
```

```

INPUT appetite
IF appetite = "irregular"
    Vata ← Vata + 1
ELSE IF appetite = "strong"
    Pitta ← Pitta + 1
ELSE IF appetite = "steady"
    Kapha ← Kapha + 1

INPUT activity
IF activity = "restless"
    Vata ← Vata + 1
ELSE IF activity = "active"
    Pitta ← Pitta + 1
ELSE IF activity = "slow"
    Kapha ← Kapha + 1

INPUT sleep
IF sleep = "light"
    Vata ← Vata + 1
ELSE IF sleep = "moderate"
    Pitta ← Pitta + 1
ELSE IF sleep = "deep"
    Kapha ← Kapha + 1

INPUT weather
IF weather = "cold"
    Vata ← Vata + 1
ELSE IF weather = "hot"
    Pitta ← Pitta + 1
ELSE IF weather = "damp"
    Kapha ← Kapha + 1

IF Vata > Pitta AND Vata > Kapha
    RESULT ← "Vata"

ELSE IF Pitta > Vata AND Pitta > Kapha
    RESULT ← "Pitta"

ELSE IF Kapha > Vata AND Kapha > Pitta
    RESULT ← "Kapha"

ELSE
    RESULT ← "Mixed / Equal Classification"

DISPLAY Vata, Pitta, Kapha
DISPLAY RESULT

STOP

```

Flow:

Input → Apply Rules → Calculate Scores → Compare Scores → Classification → Output

Working Python Code

```
# =====
# IKS CONCEPT AS CS CONCEPT
# Ayurvedic Classification - Rule-Based Expert System
# Educational Demonstration
# =====
print("=" * 60)
print(" AYURVEDIC CLASSIFICATION - RULE BASED EXPERT SYSTEM")
print("=" * 60)
# -----
# Knowledge Base
# Each option contains a rule and the corresponding dosha.
# -----
rules = {
    "body": {
        "thin": ("Vata", "Thin/light body tendency -> Vata"),
        "medium": ("Pitta", "Medium body tendency -> Pitta"),
        "solid": ("Kapha", "Solid/heavy body tendency -> Kapha")
    },
    "appetite": {
        "irregular": ("Vata", "Irregular appetite -> Vata"),
        "strong": ("Pitta", "Strong appetite -> Pitta"),
        "steady": ("Kapha", "Steady appetite -> Kapha")
    },
    "activity": {
        "restless": ("Vata", "Restless/highly variable activity -> Vata"),
        "active": ("Pitta", "Active and goal-oriented activity -> Pitta"),
        "slow": ("Kapha", "Slow/steady activity -> Kapha")
    },
    "sleep": {
        "light": ("Vata", "Light/irregular sleep -> Vata"),
        "moderate": ("Pitta", "Moderate sleep -> Pitta"),
        "deep": ("Kapha", "Deep/long sleep -> Kapha")
    },
    "weather": {
        "cold": ("Vata", "Discomfort in cold weather -> Vata"),
        "hot": ("Pitta", "Discomfort in hot weather -> Pitta"),
        "damp": ("Kapha", "Discomfort in damp weather -> Kapha")
    }
}
# -----
# Function to get valid input
# -----
def get_input(question, options):
    print("\n" + question)
    for i, option in enumerate(options, 1):
        print(f"{i}. {option}")
    while True:
        try:
```

```

        choice = int(input("Enter choice: "))
        if 1 <= choice <= len(options):
            return options[choice - 1]
        print("Please enter a valid option.")
    except ValueError:
        print("Please enter a number.")
# -----
# Collect user inputs
# -----
body = get_input(
    "Select your general body tendency:",
    ["thin", "medium", "solid"]
)
appetite = get_input(
    "Select your appetite pattern:",
    ["irregular", "strong", "steady"]
)
activity = get_input(
    "Select your activity pattern:",
    ["restless", "active", "slow"]
)
sleep = get_input(
    "Select your sleep pattern:",
    ["light", "moderate", "deep"]
)
weather = get_input(
    "Select the weather condition you generally find uncomfortable:",
    ["cold", "hot", "damp"]
)
# -----
# Initialize scores
# -----
scores = {
    "Vata": 0,
    "Pitta": 0,
    "Kapha": 0
}
matched_rules = []
# -----
# Inference Engine
# -----
inputs = {
    "body": body,
    "appetite": appetite,
    "activity": activity,
    "sleep": sleep,
    "weather": weather
}
for category, value in inputs.items():
    dosha, explanation = rules[category][value]
    scores[dosha] += 1
    matched_rules.append(
        f"IF {category} = {value} THEN {dosha}"
    )

```

```

    )
# -----
# Determine classification
# -----
highest_score = max(scores.values())
dominant_doshas = [
    dosha for dosha, score in scores.items()
    if score == highest_score
]
if len(dominant_doshas) == 1:
    result = dominant_doshas[0]
else:
    result = "Mixed / Equal Classification"
# -----
# Display Results
# -----
print("\n" + "=" * 60)
print("        EXPERT SYSTEM RESULT")
print("=" * 60)
print("\nInput Facts:")
for key, value in inputs.items():
    print(f"{key.capitalize():12}: {value}")
print("\nDosha Scores:")
for dosha, score in scores.items():
    print(f"{dosha:10}: {score}")
print("\nMatched Rules:")
for rule in matched_rules:
    print(" -", rule)
print("\n" + "-" * 60)
print("CLASSIFICATION:", result)
print("-" * 60)
if result == "Vata":
    print("Vata has the highest rule score.")
elif result == "Pitta":
    print("Pitta has the highest rule score.")
elif result == "Kapha":
    print("Kapha has the highest rule score.")
else:
    print("Two or more doshas have the same highest score.")
print("\nThis system is an educational demonstration")
print("of rule-based classification and is NOT a medical")
print("diagnosis or treatment system.")

```

Example input:

Body tendency : Thin
 Appetite : Irregular
 Activity : Restless
 Sleep : Light
 Weather : Cold

Output:

```

=====
        EXPERT SYSTEM RESULT

```

=====

Input Facts:

Body : thin
Appetite : irregular
Activity : restless
Sleep : light
Weather : cold

Dosha Scores:

Vata : 5
Pitta : 0
Kapha : 0

Matched Rules:

- IF body = thin THEN Vata
- IF appetite = irregular THEN Vata
- IF activity = restless THEN Vata
- IF sleep = light THEN Vata
- IF weather = cold THEN Vata

CLASSIFICATION: Vata

Vata has the highest rule score.
print("=" * 60)

Test Cases

The following test cases can be included in the **Ayurvedic Classification – Rule-Based Expert System** practical.

Test Case	Body Tendency	Appetite	Activity	Sleep	Weather	Expected Result
TC01	Thin	Irregular	Restless	Light	Cold	Vata
TC02	Medium	Strong	Active	Moderate	Hot	Pitta
TC03	Solid	Steady	Slow	Deep	Damp	Kapha
TC04	Thin	Irregular	Active	Light	Cold	Vata
TC05	Medium	Strong	Active	Deep	Hot	Pitta
TC06	Solid	Steady	Slow	Moderate	Damp	Kapha
TC07	Thin	Strong	Active	Moderate	Hot	Pitta
TC08	Solid	Irregular	Slow	Deep	Damp	Kapha
TC09	Medium	Irregular	Restless	Light	Cold	Vata
TC10	Medium	Strong	Slow	Moderate	Hot	Pitta

Detailed Test Case Results

Input:
Body = Thin
appetite = Irregular
Activity = Restless
Sleep = Light
Weather = Cold
Scores:
Vata = 5
Pitta = 0
Kapha = 0
Expected Output: Vata

Input:
Body = Medium
Appetite = Strong
Activity = Active
Sleep = Moderate
Weather = Hot
Scores:
Vata = 0
Pitta = 5
Kapha = 0
Expected Output: Pitta

Input:
Body = Solid
Appetite = Steady
Activity = Slow
Sleep = Deep
Weather = Damp
Scores:
Vata = 0
Pitta = 0
Kapha = 5
Expected Output: Kapha

Input:
Body = Thin
Appetite = Irregular
Activity = Active
Sleep = Light
Weather = Cold
Scores:
Vata = 3
Pitta = 2
Kapha = 0
Expected Output: Vata

Input:
Body = Medium
Appetite = Strong
Activity = Active
Sleep = Deep
Weather = Hot
Scores:
Vata = 0
Pitta = 3
Kapha = 1
Vata = 0
Expected Output: Pitta

Input:
Body = Solid
Appetite = Steady
Activity = Slow
Sleep = Moderate
Weather = Damp
Scores:
Vata = 0
Pitta = 1
Kapha = 4
Expected Output: Kapha

Input:
Body = Thin
Appetite = Strong
Activity = Active
Sleep = Moderate
Weather = Hot
Scores:
Vata = 1
Pitta = 4
Kapha = 0
Expected Output: Pitta

Input:
Body = Solid
Appetite = Irregular
Activity = Slow
Sleep = Deep
Weather = Damp
Scores:
Vata = 1
Pitta = 0
Kapha = 4
Expected Output: Kapha

Input:
Body = Medium
Appetite = Irregular
Activity = Restless
Sleep = Light
Weather = Cold
Scores:
Vata = 4
Pitta = 1
Kapha = 0
Expected Output: Vata

Input:

Body = Medium

Appetite = Strong

Activity = Slow

Sleep = Moderate

Weather = Hot

Scores:

Vata = 0

Pitta = 3

Kapha = 1

Expected Output: Pitta

Test Case Summary

Total Test Cases : 10

Vata Classification : TC01, TC04, TC09

Pitta Classification : TC02, TC05, TC07, TC10

Kapha Classification : TC03, TC06, TC08

All test cases should be executed successfully
and the actual output should match the expected result.

Complexity & Limitations

Complexity Analysis

The Ayurvedic Rule-Based Expert System checks the user's characteristics against a fixed set of rules.

Time Complexity:

$O(n)$

Where n = number of input characteristics/rules checked.

For our implementation:

- Body tendency → 1 check
- Appetite → 1 check
- Activity → 1 check
- Sleep → 1 check
- Weather → 1 check

Therefore, the system performs a small, fixed number of operations and gives the result quickly.

Space Complexity:

$O(n)$

Memory is used to store:

- Knowledge-base rules
- User inputs
- Dosha scores
- Matched rules

Limitations

1. **Simplified Knowledge:** The project uses a limited set of Ayurvedic characteristics and does not represent the complete Ayurvedic knowledge system.
2. **Fixed Rules:** The system depends on predefined IF–THEN rules and cannot automatically learn new rules.
3. **Limited Input:** Only five characteristics are considered: body tendency, appetite, activity, sleep, and weather response.

4. **Subjective Answers:** User responses may not always be accurate or consistent.
5. **No Machine Learning:** The system does not improve its classification from previous results or datasets.
6. **Mixed Classification:** When two or more doshas receive the same highest score, the system may return **Mixed/Equal Classification**.
7. **Not a Medical System:** The output is intended only for demonstrating **IKS knowledge representation and rule-based classification**. It should not be used for medical diagnosis or treatment.

Conclusion

The **Ayurvedic Classification – Rule-Based Expert System** successfully demonstrates how an **Indian Knowledge System (IKS)** can be represented using Computer Science concepts.

In this project, Ayurvedic concepts such as **Vata, Pitta, and Kapha** are represented as classification categories, while traditional characteristics are converted into **facts and IF–THEN rules**. The inference engine evaluates the user's inputs, applies the predefined rules, calculates the dosha scores, and produces a classification.

This project demonstrates important concepts of **Artificial Intelligence, knowledge representation, rule-based reasoning, inference engines, and classification**. It also shows how traditional Indian knowledge can be connected with modern computational techniques.

Therefore, the project provides a simple and practical example of converting an IKS concept into a Computer Science rule-based expert system.